mous take-off, waypoint navigation, landing precision, ground steering and complete halt can be done without any ground support," he says.

Unlike most aircraft which typically have a pair of wings, horizontal and vertical tail and fuselage, the SWiFT has just one single lift surface. "We have mastered the technology. Now it's up to the government to decide on how to go forward," says another scientist. The project is yet to get the Cabinet Committee on Security (CCS) approval. Defence scientists claim the flight test was of a scaled-down prototype while a full-scale prototype will take a couple of years to get finalised. The demonstrator is the first step towards India's own unmanned strike air vehicle.

Powered by a Russian turbofan made by NPO-Saturn, the SWiFT's airframe, undercarriage and landing gear, flight controls, and avionics system were developed indigenously. However, the majority of its avionics and electronics are derived from another drone, the Tapas-BH-201, a medium-altitude long-endurance (MALE) unmanned aerial vehicle, which is now in the final stages of development. If the Tapas proves successful, it could signal the end of the armed forces' dependence on foreign-made UAVs.

"SWiFT is much faster than other armed drones like the MQ9 Reaper or Predator or Sea Guardian. It's a stealth combat drone capable of flying with lightning speed and firing missiles and dropping bombs even before enemy radars can detect it," says a defence scientist. Defence analysts are comparing SWiFT with the US navy's X-47, French Dassault's Neuron (nEUROn) and the Russian Sukhoi S-70 Okhotnik-B.

ADE has been working on the Ghatak project, which is an autonomous jet-powered stealthy unmanned combat air vehicle. Project details are still classified, but it is believed that the Ghatak will have an internal weapons bay for carrying missiles, bombs and precision-guided munitions. Its design will be based on the flying-wing concept and will be powered by a turbofan engine.

Project Ghatak was initiated as a successor to the 2009 AURA (Autonomous Unmanned Research Aircraft) programme. AURA had a budget of Rs 12.5 crore to carry out a feasibility study for a future Indian UCAV. The ADE was already working on the Ghatak project when it was commissioned to produce the SWiFT as a prototype. Though work on the SWiFT prototype began in 2016, the project lost its momentum due to the Covid pandemic.

A majority of the stealth features in the SWiFT combat drone were developed by the Aeronautical Development Agency (ADA), a Bengaluru-based aircraft design and development establishment working on India's own 6th Generation Stealth Advanced Medium Combat Aircraft (AMCA). The stealth technology was transferred by ADA to scientists at the ADE.

India is among the few modern militaries that still do not have armed drones. Even smaller countries like Pakistan, Nigeria, Somalia and South Africa are now using weaponised drones.

Currently, all three wings of the armed forces are dependent on foreign-made drones like the Israel-made Heron and Searchers, and that too only to gather intelligence. That said, under Project Cheetah, the Indian Air Force (IAF) is looking to upgrade its existing fleet of Heron drones for offensive missions. The medium-altitude, long-endurance Israeli drones are being fitted with laser-guided bombs and air-to-ground and air-launched anti-tank missiles. The air force, which is the lead agency for the project, will spend Rs 5,000 crore on the upgrades.

The war in Ukraine has once again highlighted the role of drones in combat. Video footage circulating on the internet showed how the Turkish combat drone Bayraktar TB2 successfully carried out strikes against the Russian army and its armoured vehicles. The same Bayraktar was used in Libya and in the battle between Azerbaijan and Armenia over Nagorno-Karabakh in 2020. Videos of Armenian tanks and artillery positions being decimated by Azerbaijan's drones showed the world the changing character of new-age wars. ■

GHATAK DRONE

STEALTH

Stealth tech to avoid radar detection, 70% of which will come from the design, 30% from material/ tech such as radar-absorbent paint, radio frequency reduction techniques

INTERNAL WEAPONS BAY

Can carry a minimum of 2 laser-guided bombs

WEIGHT

15 tonnes (max. take-off weight, including fuel and weapons payload)

Many of the stealth features in SWiFT were developed by ADA, which is working on India's own 6th Gen Stealth combat aircraft